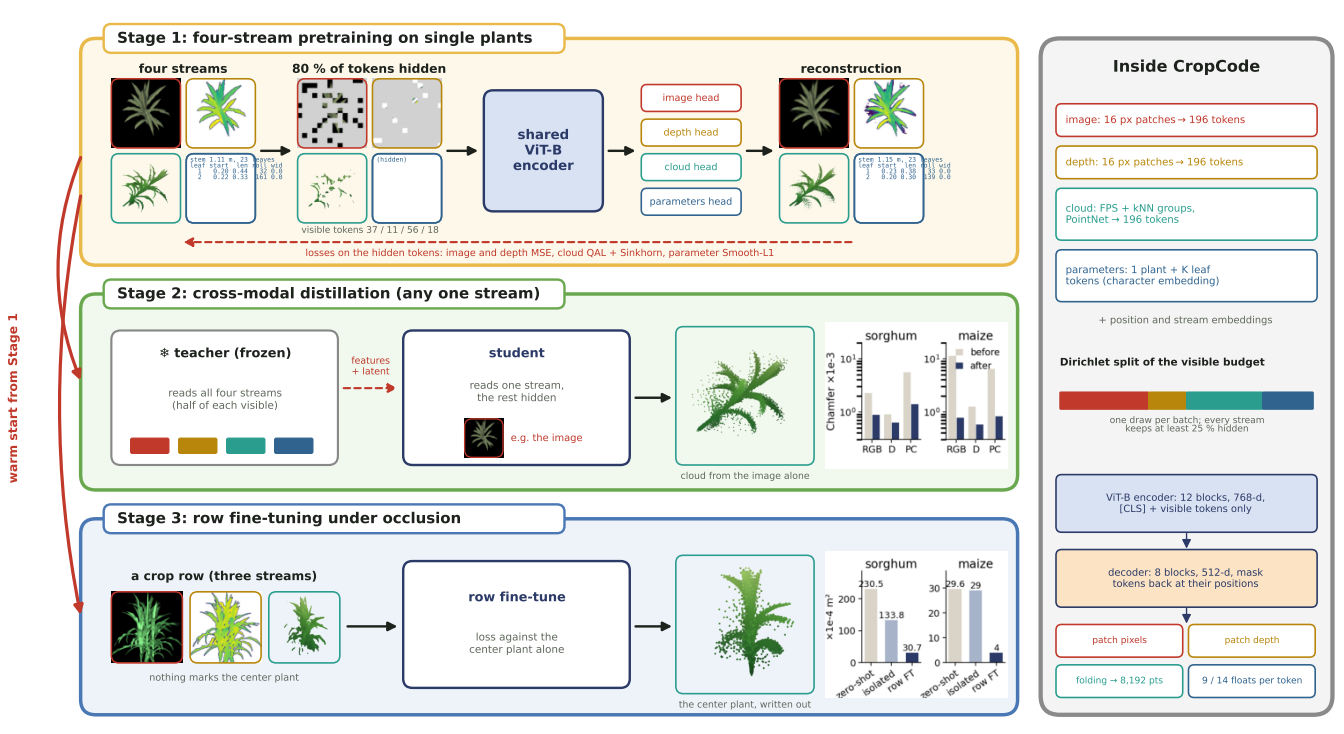

Stage 1: four-stream pretraining on single plants
four streams
80 % of tokens hidden
shared ViT-B encoder
image head
depth head
cloud head
parameters head
reconstruction
visible tokens 37 / 11 / 56 / 18
losses on the hidden tokens: image and depth MSE, cloud QAL + Sinkhorn, parameter Smooth-L1
warm start from Stage 1
Stage 2: cross-modal distillation (any one stream)
❄ teacher (frozen)
reads all four streams
(half of each visible)
features + latent
student
reads one stream, the rest hidden
e.g. the image
cloud from the image alone
sorghum
maize
before
after
Chamfer ×1e-3
RGB D PC
Stage 3: row fine-tuning under occlusion
a crop row (three streams)
nothing marks the center plant
row fine-tune
loss against the center plant alone
the center plant, written out
×1e-4 m²
230.5
133.8
30.7
29.6
29
4
zero-shot
isolated
row FT
Inside CropCode
image: 16 px patches → 196 tokens
depth: 16 px patches → 196 tokens
cloud: FPS + kNN groups, PointNet → 196 tokens
parameters: 1 plant + K leaf tokens (character embedding)
+ position and stream embeddings
Dirichlet split of the visible budget
one draw per batch; every stream keeps at least 25 % hidden
ViT-B encoder: 12 blocks, 768-d, [CLS] + visible tokens only
decoder: 8 blocks, 512-d, mask tokens back at their positions
patch pixels
patch depth
folding → 8,192 pts
9 / 14 floats per token